



**Politecnico
di Torino**

Visual Semantic Reasoning for Image-Text Matching

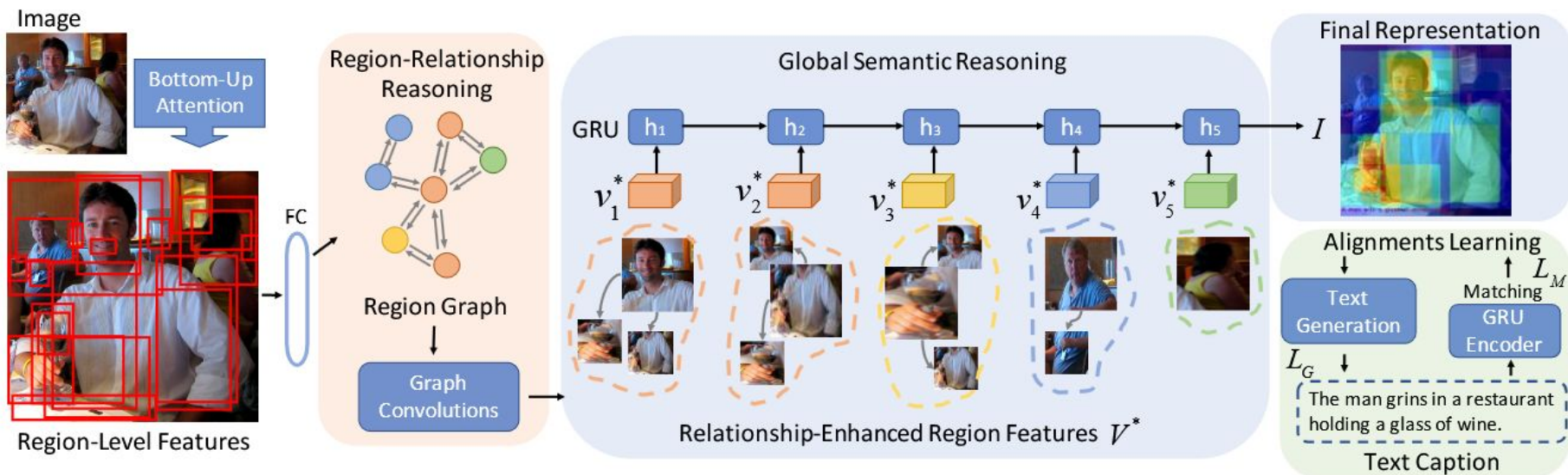
by Kunpeng Li et al.

Published in 2019 at ICCV (International Conference on Computer Vision) Seoul, Korea

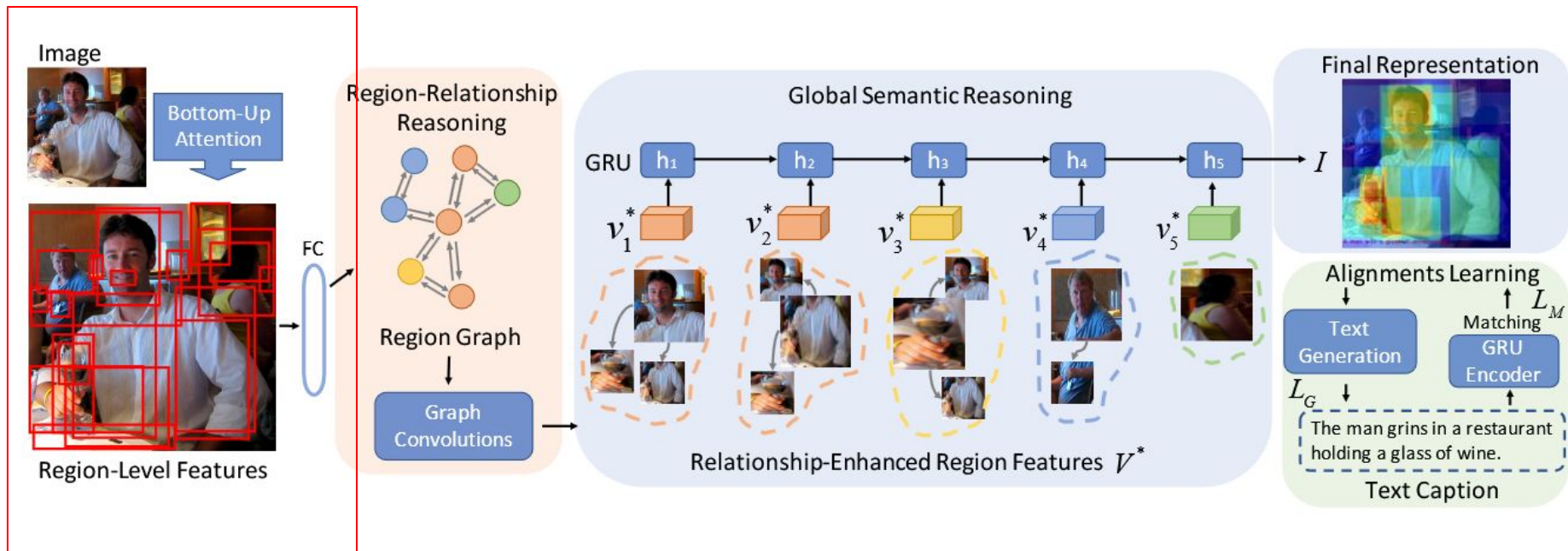
Presented by:

Luca Datti, Francesco Martino, Maura Gallinari

The Pipeline: An Overview



Visual Exploration Stage





Vanilla CNNs: Some Drawbacks



Fixed Receptive Fields

Narrow perspective on image exploration



Lack of Semantic Context

Inability to understand contextual relationship between objects:

E.g. Person + laptop + mug = work



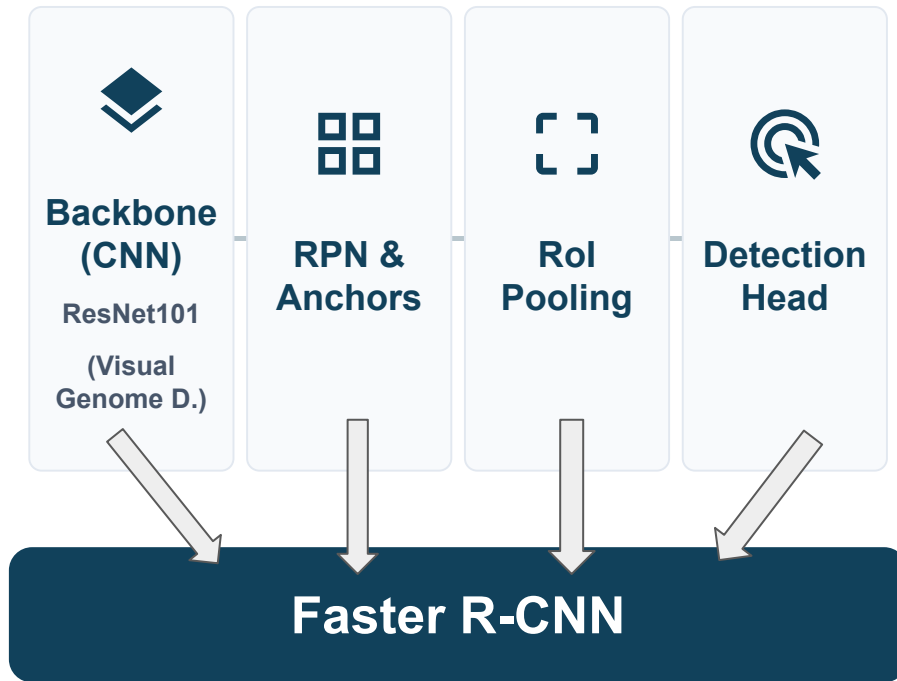
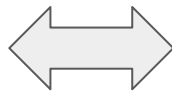
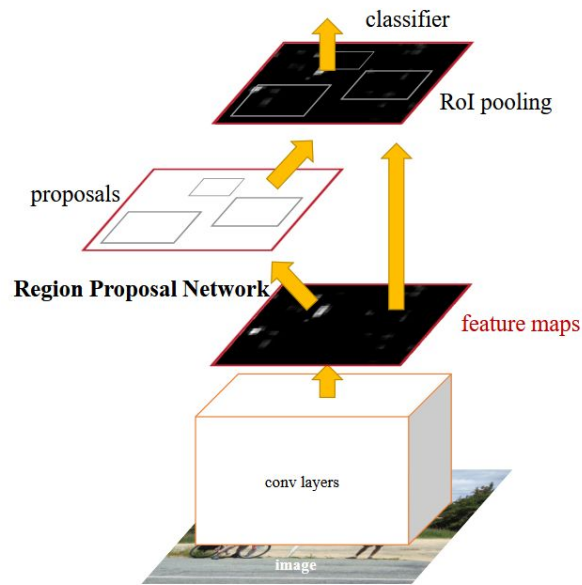
Lack of Relational Logic

Can't grasp objects interaction

E.g. Person + pen = writing



Adding Complexity: Faster R-CNN

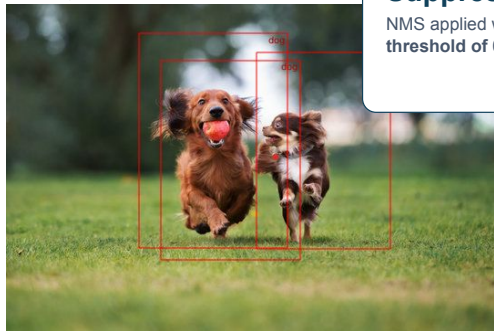


RoI Selection



1)

How many dogs are there?



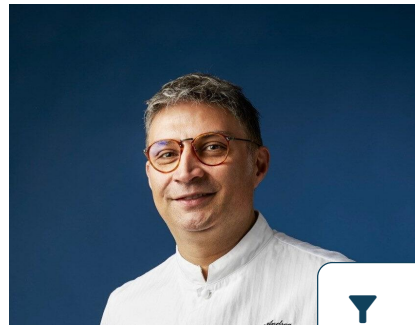
Source: https://stock.adobe.com/it/search?k=dogs+playing&asset_id=121096965



Redundancy Suppression

NMS applied with a strict IoU threshold of 0.7.

2)



Source:
<https://ambasciatoridelgusto.it/portfolio/andre-a-medici/>

Is it a doctor or a chef?

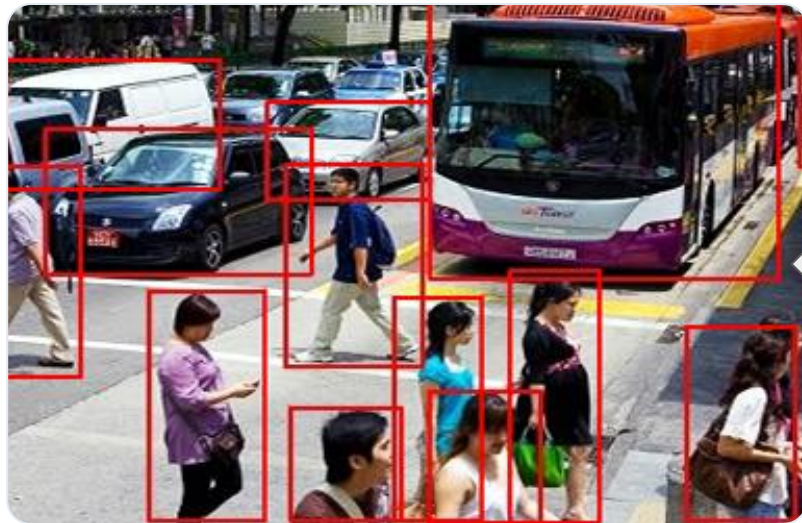


Confidence Filtering

Robust detection threshold set at 0.3

3) **Top-36** regions selection (by confidence)

One Image, Many Embeddings



Source: https://www.analytics.ai/public/upload/1697088147_Importance-of-Image-Annotation-in-Machine-Learning.jpg

Feature Extraction & Embedding

1. Extraction

Features are extracted for each region i via average pooling, producing a feature vector f_i with 2048 dimensions.

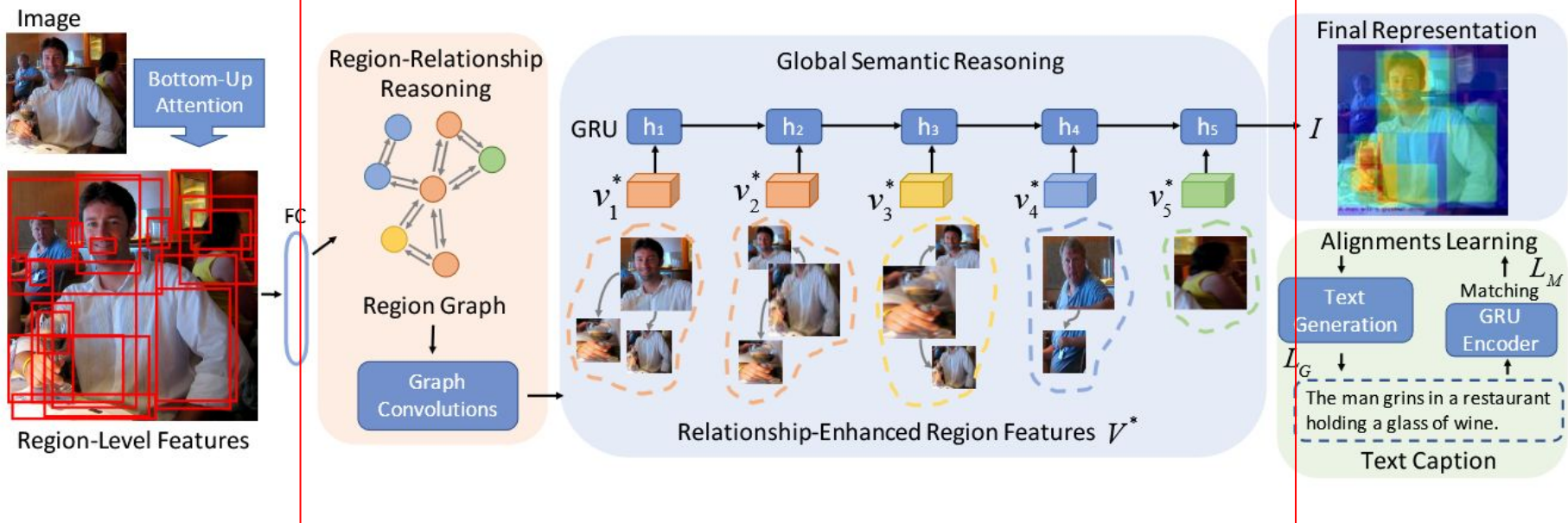
2. Transformation

Fully-connected layer projects f_i to a lower-dimensional space.

$$\psi : \mathbb{R}^{2048} \rightarrow \mathbb{R}^D$$
$$f_i \mapsto v_i = W_f f_i + b_f$$

W_f weight matrix,
 b_f bias vector

Reasoning stage

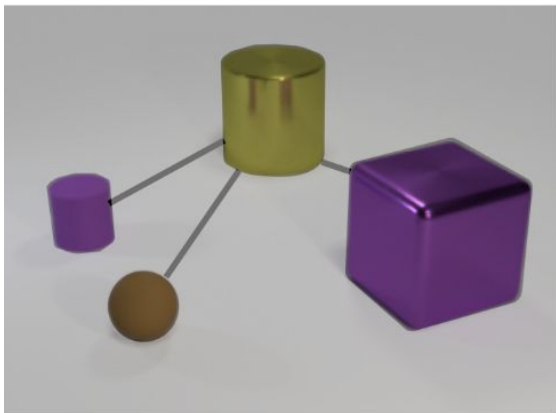




Semantic Correlation Reasoning

Pairwise Affinity Computation

- **Backpropagation Learning:** optimizes W_φ and W_ϕ to learn semantic dependencies.
- **Relationship Construction:** measures affinity and captures semantic relationships among detected image regions.



$$\varphi(v_i) = W_\varphi v_i$$

$$\phi(v_j) = W_\phi v_j$$

$$R(v_i, v_j) = \varphi(v_i)^T \phi(v_j)$$

Affinity Matrix (R): Pairwise measure of semantic correlation.

GCN For Global Semantic Reasoning



Connected Topology

High affinity scores generate strong edges between regions that are highly correlated.

GCN Reasoning

Information is propagated across the graph; node responses are computed based on their weighted semantic neighbors.

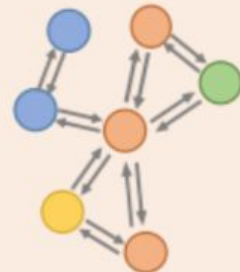
Residual GCN Formula

We add residual connections to the original GCN as follows:

$$V^* = W_r(RVW_g) + V$$

W_r: Residual Weight Matrix. **R**: Affinity Matrix. **V**: Original Input Matrix. **W_g**: GCN Weight Matrix.

Region-Relationship Reasoning

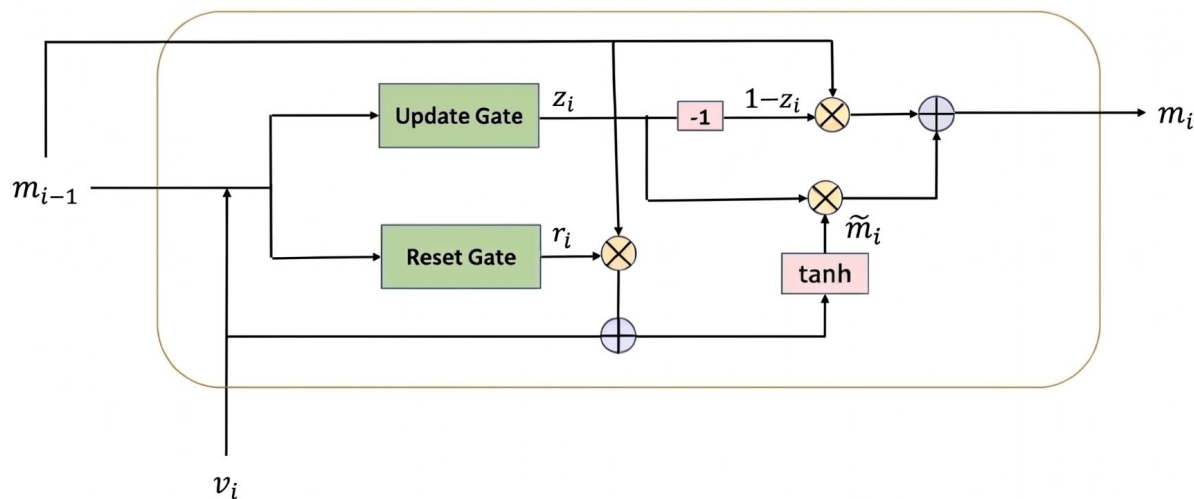


$Gr = (V, E)$

V: detected regions

E: affinity matrix R scores

Gated Recurrent Unit (GRU) Architecture



Key Components

- **Reset Gate**

$$r_i = \sigma_r(W_r v_i^* + U_r m_{i-1} + b_r)$$
- **Possible candidate**

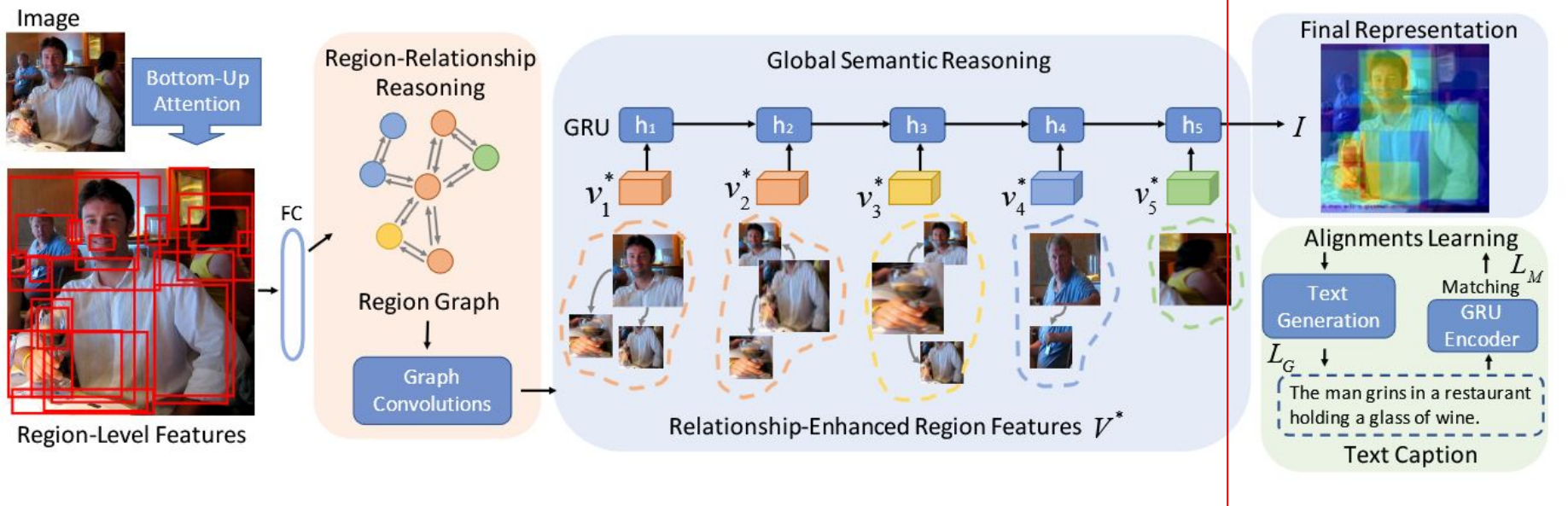
$$\tilde{m}_i = \sigma_m(W_m v_i^* + U_z(r_i \circ m_{i-1}) + b_m)$$
- **Update Gate**

$$z_i = \sigma_z(W_z v_i^* + U_z m_{i-1} + b_z)$$
- **Global memory**

$$m_i = (1 - z_i) \circ m_{i-1} + z_i \circ \tilde{m}_i$$

$W_z, W_r, W_m \in \mathbb{R}^{D \times D}$ and $U_z, U_r, U_m \in \mathbb{R}^{D \times D}$ are **weight matrices**.
 $b_z, b_r, b_m \in \mathbb{R}^D$ are **bias vectors**.

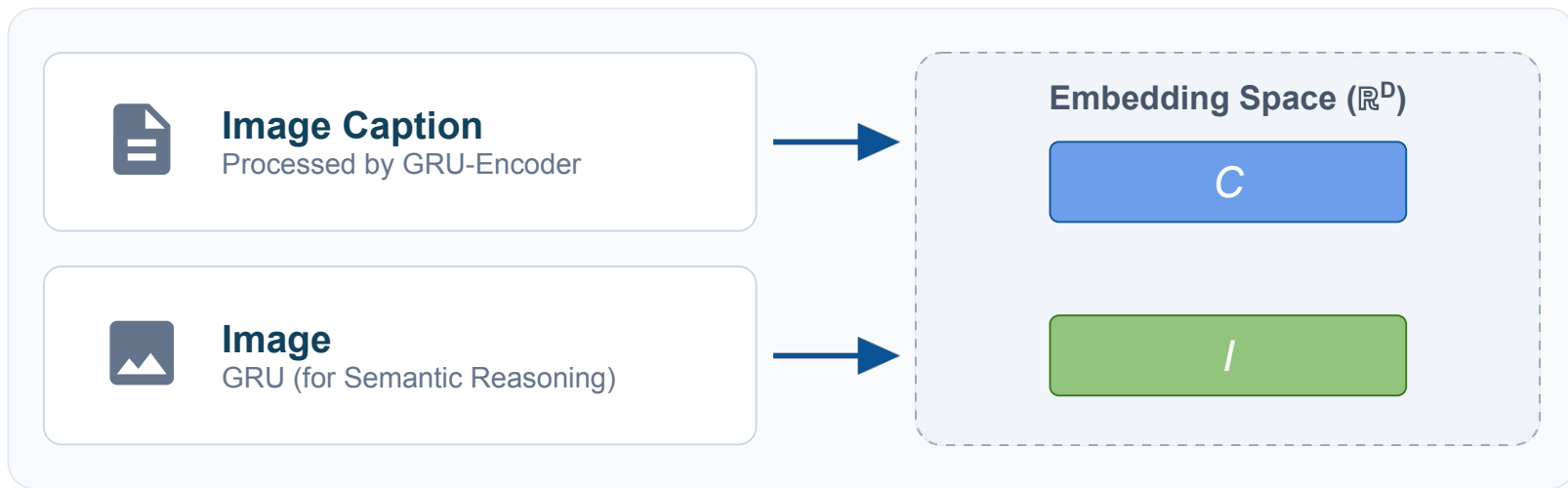
Image-Text Matching Stage





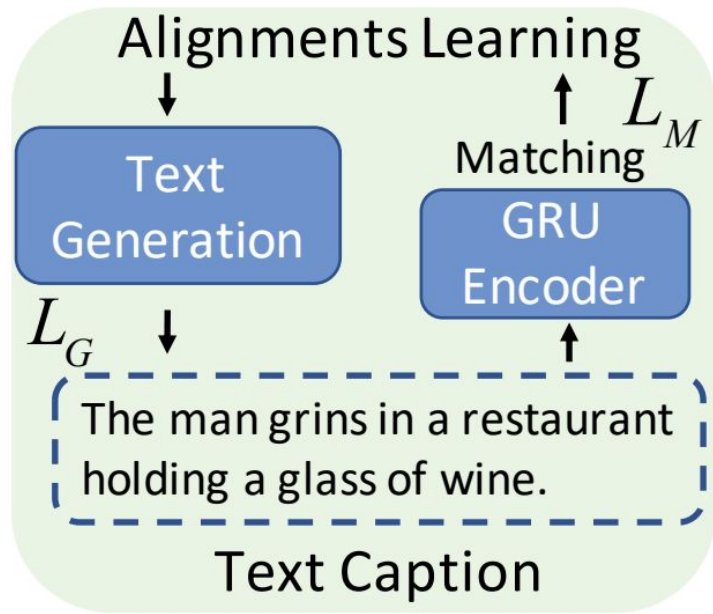
Moving to a Common Domain

We use **GRU-based encoders** to **transpose** visual and textual representations into a common **embedding space**.





Aligning Vision To Language



Two tasks (two losses):

Text-to-image Matching
(Matching Loss L_M)

Image-to-text Generation
(Generation Loss L_G)

$$L = L_M + L_G$$

Matching Loss



Loss reduction strategy

hardest negatives are kept at a fixed distance α from true positives to reduce mismatching

$$L_M = [\alpha - S(I, C) + S(I, \hat{C})]_+ + [\alpha - S(I, C) + S(\hat{I}, C)]_+,$$

where:

- α is an hyper-parameter
- I is a ground-truth image
- C is a ground-truth caption
- $\hat{I}$ is images' hardest negative (per batch)
- $\hat{C}$ is captions' hardest negative (per batch)
- $S(\cdot)$ is a similarity function

Generation Loss



Loss reduction strategy

Given an image representation V^* , it is fed to a Seq2Seq Encoder, producing a textual caption $(y_1, \dots, y_l)$.

Transposition accuracy is optimized through a sequence-to-sequence model parameter θ .

$$L_G = - \sum_{t=1}^l \log p(y_t | y_{t-1}, V^*; \theta)$$

where:

- $y_1, \dots, y_l$ is the textual caption
- V^* is the image representation
- θ is the encoder parameter (to be learned)

Experimental Setup



Benchmarks:

MS-COCO: 123k images, 5 captions each (113k train / 5k val / 5k test).

Flickr30K: 31k images, 5 captions each (28k train / 1k val / 1k test).

Evaluation Metrics:

Recall@K (R@1, R@5, R@10).

Hardware & Visual Backbone:

Faster R-CNN with ResNet-101 (pre-trained on Visual Genome).

Hyperparameters:

Adam Optimizer, 30 epochs, mini-batch size 128.

Alpha = 0.2.

Embedding space dimension D = 2048.



Quantitative Results

Improvements on Flickr30K:

- +12.6% in Image Retrieval
- +5.8% in Caption Retrieval.

Methods	Caption Retrieval			Image Retrieval		
	R@1	R@5	R@10	R@1	R@5	R@10
(R-CNN, AlexNet)						
DVSA _{CVPR'15} [15]	22.2	48.2	61.4	15.2	37.7	50.5
HMIstm _{ICCV'17} [33]	38.1	-	76.5	27.7	-	68.8
(VGG)						
FV _{CVPR'15} [20]	35.0	62.0	73.8	25.0	52.7	66.0
VQA _{ECCV'16} [29]	33.9	62.5	74.5	24.9	52.6	64.8
SMIstm _{CVPR'17} [13]	42.5	71.9	81.5	30.2	60.4	72.3
2WayN _{CVPR'17} [4]	49.8	67.5	-	36.0	55.6	-
(ResNet)						
RRF _{ICCV'17} [30]	47.6	77.4	87.1	35.4	68.3	79.9
VSE++ _{BMVC'18} [5]	52.9	79.1	87.2	39.6	69.6	79.5
SCO _{CVPR'18} [14]	55.5	82.0	89.3	41.1	70.5	80.1
(Faster R-CNN, ResNet)						
SCAN _{ECCV'18} [23]	67.4	90.3	95.8	48.6	77.7	85.2
VSRN (ours)	71.3	90.6	96.0	54.7	81.8	88.2

Table 3. Quantitative evaluation results of the image-to-text (caption) retrieval and text-to-image (image) retrieval on Flickr30K test set in terms of Recall@K (R@K).

Improvements on MS-COCO:

- +6.8% in Image Retrieval
- +4.8% in Caption Retrieval.

Methods	Caption Retrieval			Image Retrieval		
	R@1	R@5	R@10	R@1	R@5	R@10
(R-CNN, AlexNet)						
DVSA _{CVPR'15} [15]	38.4	69.9	80.5	27.4	60.2	74.8
HMIstm _{ICCV'17} [33]	43.9	-	87.8	36.1	-	86.7
(VGG)						
FV _{CVPR'15} [20]	39.4	67.9	80.9	25.1	59.8	76.6
OEM _{ICLR'16} [36]	46.7	-	88.9	37.9	-	85.9
VQA _{ECCV'16} [29]	50.5	80.1	89.7	37.0	70.9	82.9
SMIstm _{CVPR'17} [13]	53.2	83.1	91.5	40.7	75.8	87.4
2WayN _{CVPR'17} [4]	55.8	75.2	-	39.7	63.3	-
(ResNet)						
RRF _{ICCV'17} [30]	56.4	85.3	91.5	43.9	78.1	88.6
VSE++ _{BMVC'18} [5]	64.6	89.1	95.7	52.0	83.1	92.0
GXN _{CVPR'18} [8]	68.5	-	97.9	56.6	-	94.5
SCO _{CVPR'18} [14]	69.9	92.9	97.5	56.7	87.5	94.8
(Faster R-CNN, ResNet)						
SCAN _{ECCV'18} [23]	72.7	94.8	98.4	58.8	88.4	94.8
VSRN (ours)	76.2	94.8	98.2	62.8	89.7	95.1

Table 1. Quantitative evaluation results of the image-to-text (caption) retrieval and text-to-image (image) retrieval on MS-COCO 1K test set in terms of Recall@K (R@K).

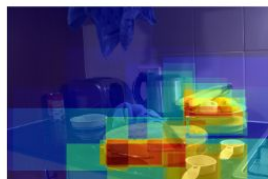
Attention Visualization



(a) Query Image

Attention

- 1: A woman and child playing frisbee in a grassy area. ✓
- 2: A woman and a little girl playing with a Frisbee in the sun on a green lawn. ✓
- 3: A woman throwing a frisbee as a child looks on. ✓



(d) Query Image

Attention

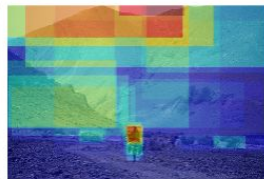
- 1: On a kitchen counter top, a knife rests on an empty cutting board. ✓
- 2: Dirty kitchen utensils and a stove together with fruit. ✓
- 3: There are various food items on a kitchen counter. ✗



(b) Query Image

Attention

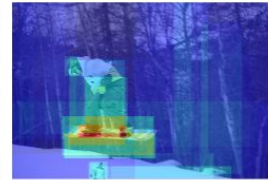
- 1: Two giraffes standing near trees in a grassy area. ✓
- 2: Two giraffes standing next to each other on a grassy field. ✓
- 3: Two giraffes rub their necks together as they stand by the trees in the sunlight. ✓



(e) Query Image

Attention

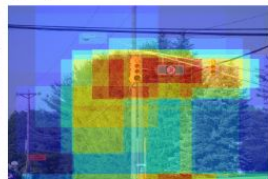
- 1: A person riding a donkey travels between two mountains. ✓
- 2: A person in a black top riding a horse and some hills and rocks. ✓
- 3: A person riding a horse down a trail with rocks around it. ✓



(c) Query Image

Attention

- 1: A man in a helmet jumps a snowboard. ✓
- 2: A snowboarder in mid-air with another person watching in the background. ✓
- 3: A person on a snowboard jumping in the air. ✓



(f) Query Image

Attention

- 1: Traffic light hanging near power lines with trees in back. ✓
- 2: Two traffic signals on a pole arm at an intersection. ✓
- 3: A street sign and two traffic lights hang over US Route 1 ✗

Conclusions



Visual Semantic Reasoning

Simple &
Interpretable

Alignment

Global > Local

New SOTA

Conclusions



Expanded Version: VSRN++

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Image-Text Embedding Learning via Visual and Textual Semantic Reasoning

Kunpeng Li^{id}, *Member, IEEE*, Yulun Zhang^{id}, *Member, IEEE*, Kai Li, *Member, IEEE*,
Yuanyuan Li^{id}, *Student Member, IEEE*, and Yun Fu^{id}, *Fellow, IEEE*

Global > Local

~~New SOTA~~



**Politecnico
di Torino**

Thank you for your attention.

Any questions?

Visual Genome Dataset



108K
IMAGES

35

Avg. Objects / Image

26

Attributes

21

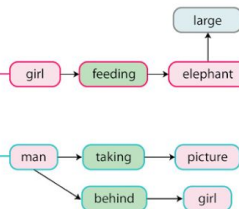
Pairwise Relationships

To produce

Scene Graph



flap flops on the ground
Elephant that could carry people
Leaves on the ground
Huts on a hillside
A bag
A bush next to a river.
a woman wearing a brown shirt
Girl feeding large elephant
Woman wearing a purple dress
Tree near the water
a man wearing a hat
A handle of bananas.
a man taking a picture behind girl
Glasses on the hair.
blue flip flop sandals
small houses on the hillside
the nearby river
Elephant with carrier on it's back
Two people near elephants



Fonte: <http://vision.stanford.edu/pdf/visualgenome.pdf>

Ablation Studies

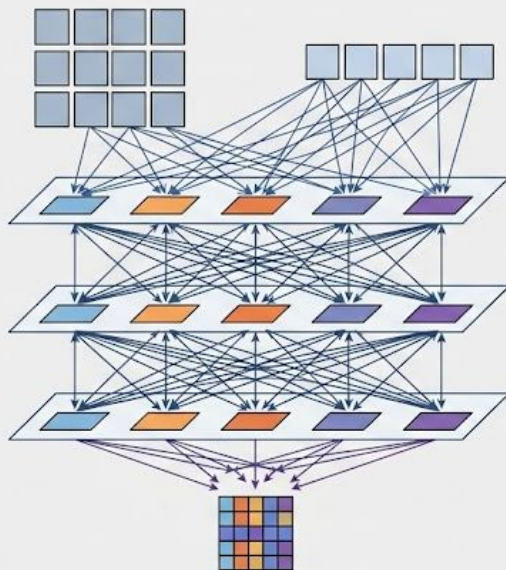


Methods	Caption Retrieval			Image Retrieval		
	R@1	R@5	R@10	R@1	R@5	R@10
Mean-pool	64.3	90.5	95.1	49.2	83.4	91.5
RRR	68.5	93.2	96.3	56.8	87.2	94.2
GSR	72.3	94.4	98.0	59.6	88.6	94.5
1RRR + GSR	75.3	94.7	98.1	62.1	89.2	94.9
4RRR + GSR	76.2	94.8	98.2	62.8	89.7	95.1
4RRR + GSR*	74.6	94.6	98.2	61.2	89.0	94.8

SCAN vs VSRN

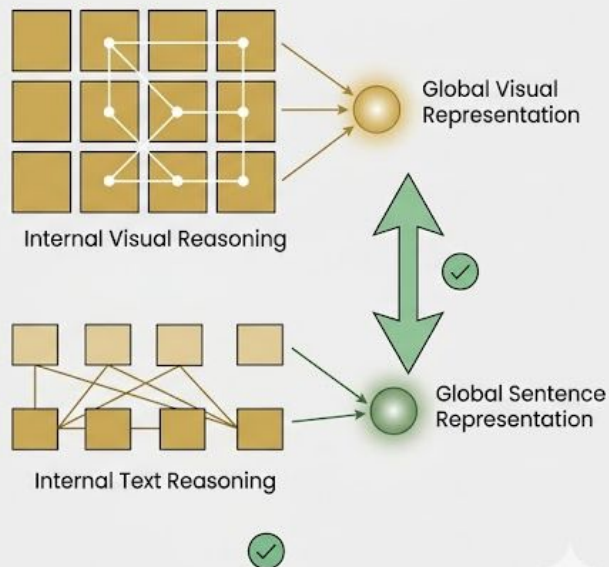


STACKED CROSS ATTENTION (SCAN)



Dense Local Cross-Modal Interactions

VISUAL SEMANTIC REASONING (VSRN)



Global Sparse Alignment